

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('ggplot')
```

```
In [2]: df = pd.read_csv('traffic.csv')

df.head()
df.info()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 226278 entries, 0 to 226277
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0   event       226278 non-null  object
1   date        226278 non-null  object
2   country     226267 non-null  object
3   city        226267 non-null  object
4   artist      226241 non-null  object
5   album       226273 non-null  object
6   track       226273 non-null  object
7   isrc        219157 non-null  object
8   linkid      226278 non-null  object
dtypes: object(9)
memory usage: 15.5+ MB
```

```
Out[2]: event      0
date      0
country    11
city      11
artist     37
album      5
track      5
isrc      7121
linkid     0
dtype: int64
```

```
In [3]: # Check duplicates
df.duplicated().sum()

# Remove duplicates
df = df.drop_duplicates()

# Missing values
df.isnull().sum()

# Fill missing values
df['country'] = df['country'].fillna('Unknown')
df['city'] = df['city'].fillna('Unknown')
df['artist'] = df['artist'].fillna('Unknown')
df['album'] = df['album'].fillna('Unknown')
df['track'] = df['track'].fillna('Unknown')
df['isrc'] = df['isrc'].fillna('Unknown')

# Convert date
```

```
df['date'] = pd.to_datetime(df['date'])

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 122567 entries, 0 to 226274
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0   event       122567 non-null  object
1   date        122567 non-null  datetime64[ns]
2   country     122567 non-null  object
3   city        122567 non-null  object
4   artist      122567 non-null  object
5   album       122567 non-null  object
6   track       122567 non-null  object
7   isrc        122567 non-null  object
8   linkid      122567 non-null  object
dtypes: datetime64[ns](1), object(8)
memory usage: 9.4+ MB
```

```
In [4]: print("Total Events:", len(df))
```

Total Events: 122567

```
In [5]: print("Unique Links:", df['linkid'].nunique())
```

Unique Links: 3839

```
In [6]: df['event'].value_counts()
```

```
Out[6]: event
pageview    73360
click       32499
preview     16708
Name: count, dtype: int64
```

```
In [7]: df['country'].value_counts().head(10)
```

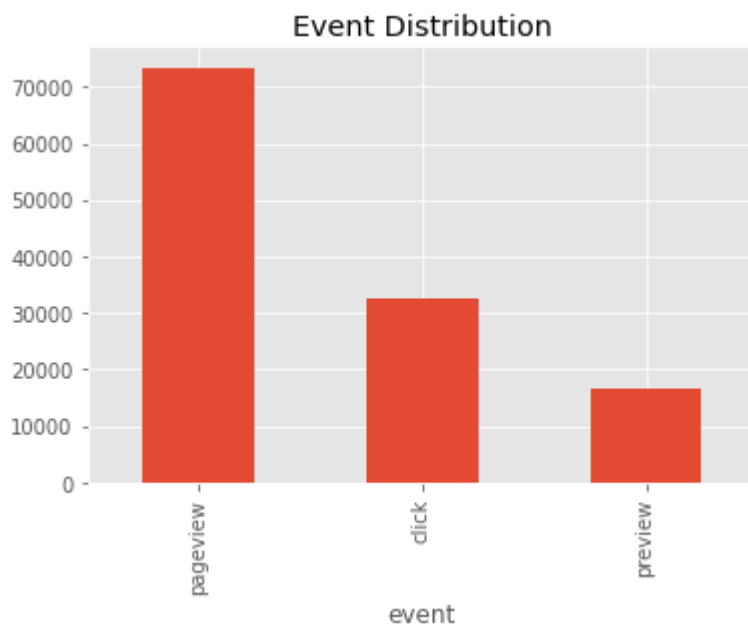
```
Out[7]: country
United States    28664
India            18689
France           10565
Saudi Arabia     7682
United Kingdom   5095
Germany          4017
Canada           2784
Pakistan         2633
Iraq             2444
Turkey           2399
Name: count, dtype: int64
```

```
In [8]: df['artist'].value_counts().head(10)
```

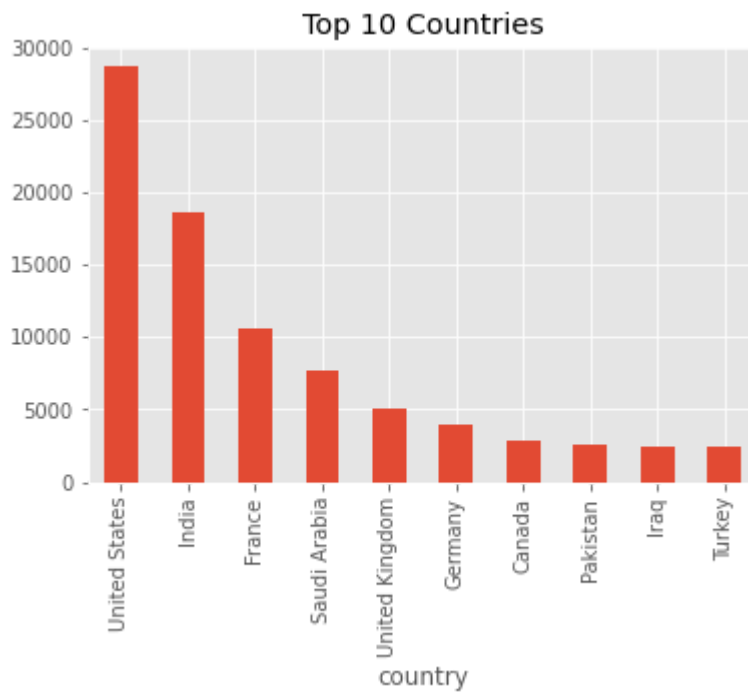
```
Out[8]: artist
Teshar 8288
Anne-Marie 4029
Tundra Beats 3951
Roddy Ricch 3107
Olivia Rodrigo 3037
Surf Mesa, Emilee 2956
DMNDS, Strange Fruits Music, Fallen Roses, Lujavo, Nito-Onna 2865
Reyanna Maria 2675
PinkPantheress 2446
50 Cent, Olivia 2390
Name: count, dtype: int64
```

```
In [9]: import matplotlib.pyplot as plt

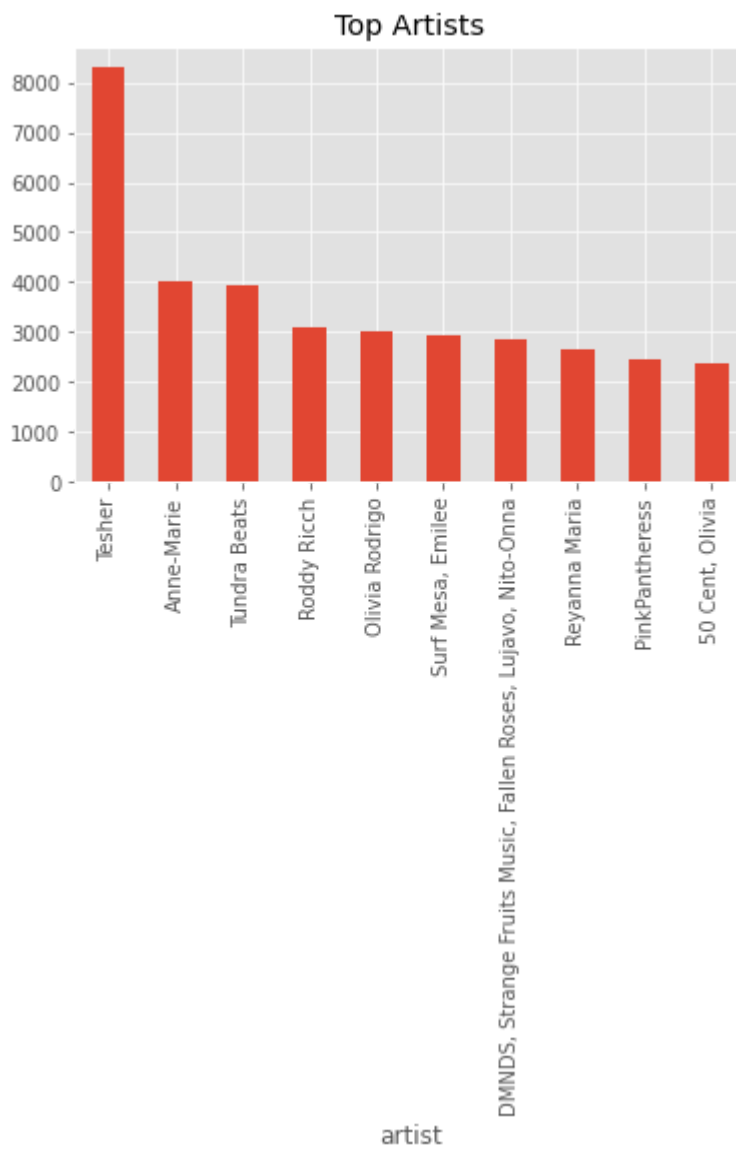
df['event'].value_counts().plot(kind='bar')
plt.title('Event Distribution')
plt.show()
```



```
In [10]: df['country'].value_counts().head(10).plot(kind='bar')
plt.title('Top 10 Countries')
plt.show()
```



```
In [11]: df['artist'].value_counts().head(10).plot(kind='bar')
plt.title('Top Artists')
plt.show()
```



```
In [ ]: 5. Recommendations for Alfido Tech
        1.Improve pages with high pageviews but low clicks.
        2.Promote top-performing artists/content more prominently.
        3. Optimize content for high-traffic countries.
        4. Use A/B testing to improve click-through rate.
        5. Analyze preview-to-click conversion and improve user journey.
```